Appendix D - Beagle Marine Park Data Analysis Report (stereo BRUV)

South-east Marine Parks Network

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Data Analysis

Statistical methods and model selection results underpinning the natural values reported in Appendix A.

Benthic ecosystem extent

Table D.1: Habitat model selection results. Top generalised additive models (GAMs) for predicting the probability of occurrence of key habitats from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the most parsimonious model (fewest variables - shown in bold) within two units of the lowest AICc.

Response	Model	Δ AICc	ω AICc	R^2	EDF
Sand	**year + geoscience_roughness + geoscience_aspect(year) + geoscience_detrended(year)**	**0**	**0.423**	**0.05575**	**10.45**
	year + geoscience_aspect(year) + geoscience_detrended(year) + geoscience_roughness(year)	0.878	0.273	0.05641	11.75
	year + geoscience_detrended + geoscience_aspect(year) + geoscience_roughness(year)	1.238	0.228	0.05511	10.19
Sessile invertebrates	**year + geoscience_roughness + geoscience_aspect(year) + geoscience_detrended(year)**	**0**	**0.467**	**0.0504**	**10.44**
	year + geoscience_aspect(year) + geoscience_detrended(year) + geoscience_roughness(year)	1.507	0.22	0.05088	11.75
	year + geoscience_detrended + geoscience_aspect(year) + geoscience_roughness(year)	1.909	0.18	0.04949	10.19
Reef	**year + geoscience_roughness + geoscience_aspect(year) + geoscience_detrended(year)**	**0**	**0.57**	**0.05263**	**10.46**
	year + geoscience_aspect(year) + geoscience_detrended(year) + geoscience_roughness(year)	1.682	0.246	0.05301	11.74

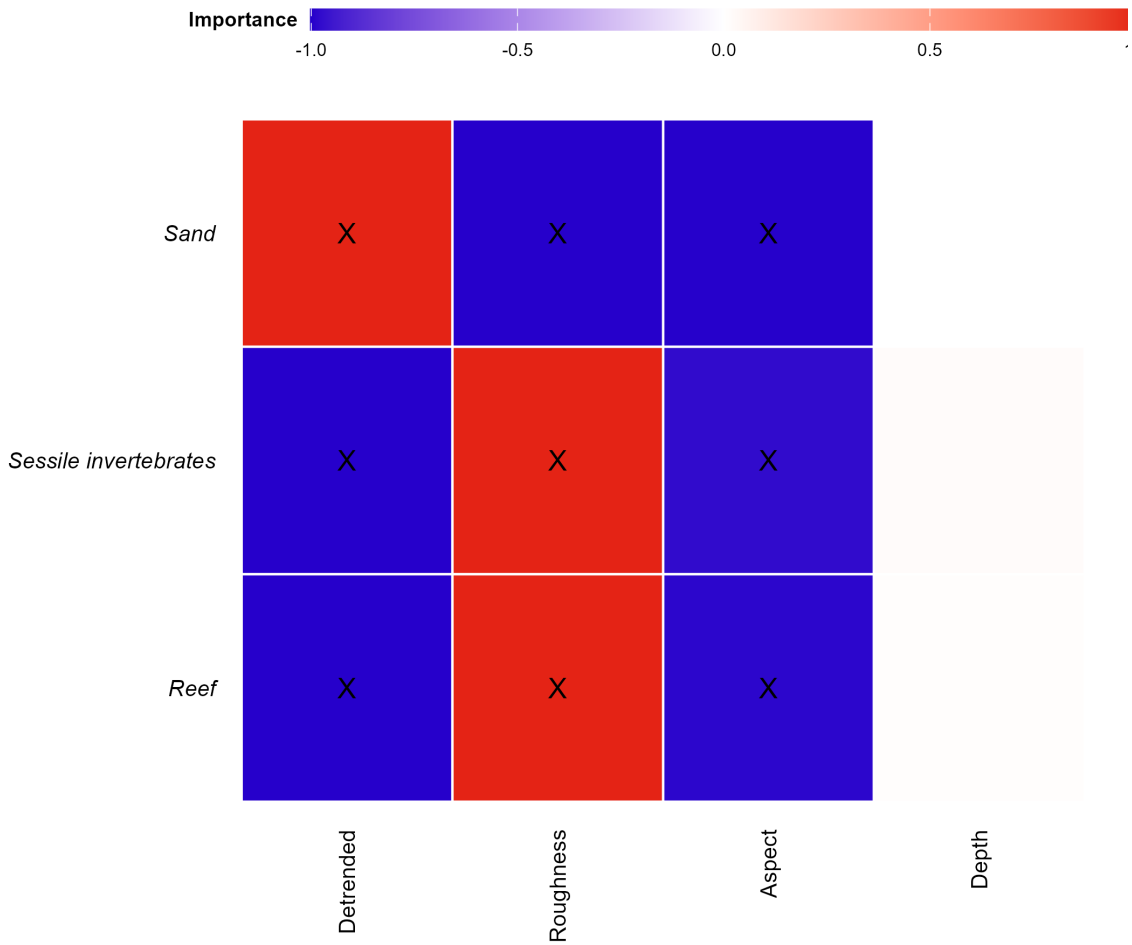


Figure D.1: Variable importance scores for predicting the probability of occurrence of main habitat categories. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient.

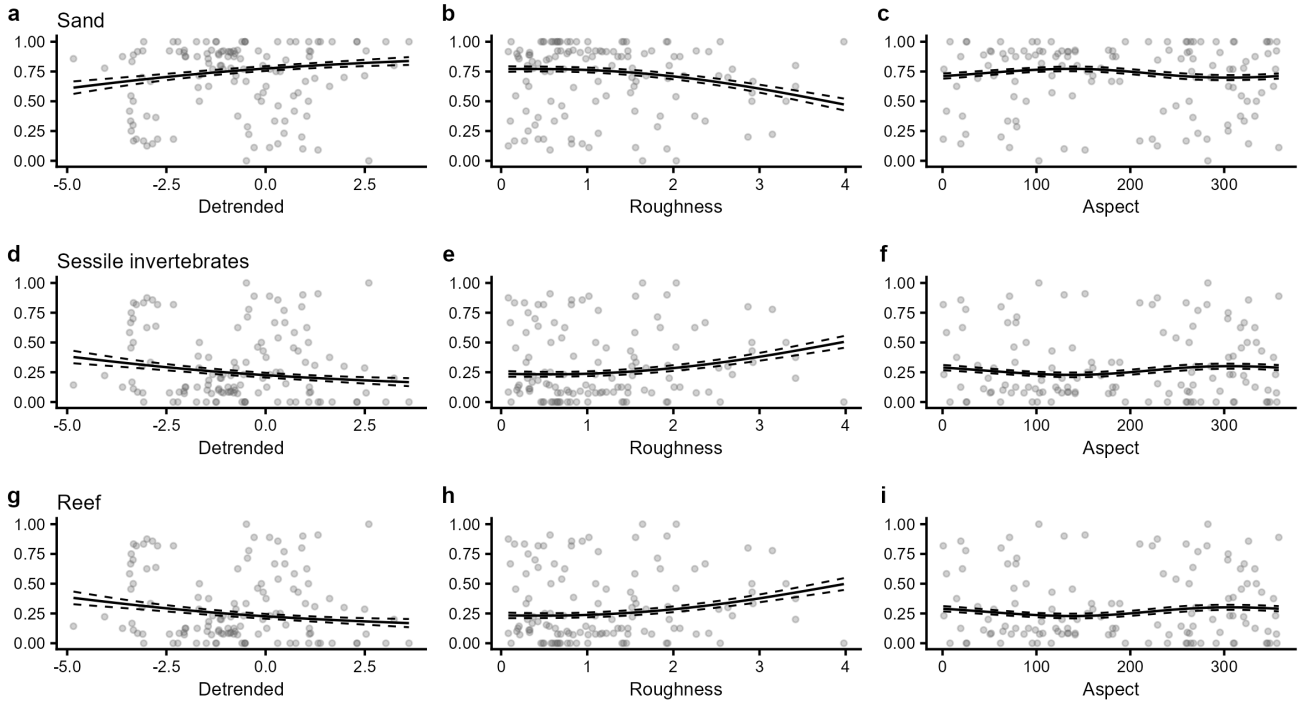


Figure D.2: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years.

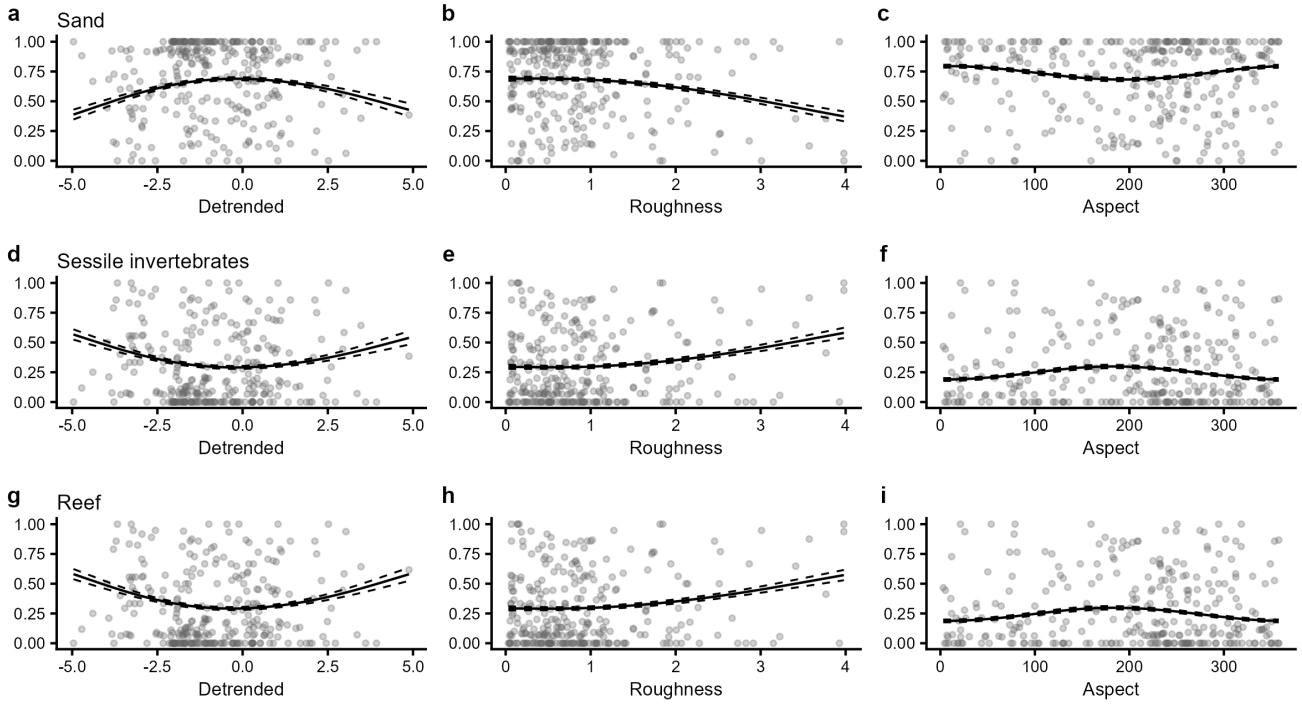


Figure D.3: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years.

Demersal fish assemblage

Table D.2: Demersal fish model selection results. Top generalised additive models (GAMs) for predicting the species and abundance distribution of metrics of interest from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the best model (fewest variables - shown in bold) within two units of the lowest AICc. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Response	Model	Δ AICc	ω AICc	AICc	R^2	EDF
Species richness	**detrended + aspect(year) + Z(year) + reef + year**	**0**	**0.105**	**2191.81**	**0.27643**	**9.62**
	detrended + Z(year) + reef + year	0.203	0.095	2192.01	0.27026	7.48
	detrended + aspect + Z(year) + reef + year	0.252	0.093	2192.06	0.27035	8.48
	detrended + aspect(year) + Z(year) + reef(year) + year	0.738	0.073	2192.55	0.27979	10.93
	detrended(year) + aspect(year) + Z(year) + reef + year	0.897	0.067	2192.71	0.27795	10.52
	detrended(year) + Z(year) + reef + year	1.064	0.062	2192.87	0.27171	8.37
	detrended + Z(year) + reef(year) + year	1.226	0.057	2193.03	0.27309	8.77
	detrended(year) + aspect + Z(year) + reef + year	1.296	0.055	2193.10	0.27216	9.38
	detrended + aspect + Z(year) + reef(year) + year	1.487	0.05	2193.30	0.27362	9.77
	detrended(year) + aspect(year) + Z(year) + reef(year) + year	1.552	0.048	2193.36	0.28146	11.85
Total abundance	**detrended(year) + Z(year) + reef(year) + year**	**0**	**0.127**	**3609.90**	**0.42501**	**11.34**
	detrended(year) + aspect(year) + Z(year) + reef(year) + year	0.001	0.127	3609.90	0.42502	13.33
	detrended(year) + aspect + Z(year) + reef(year) + year	0.004	0.126	3609.90	0.42502	12.34
	Z(year) + reef(year) + year	0.912	0.08	3610.81	0.41611	8.64
	aspect(year) + Z(year) + reef(year) + year	0.912	0.08	3610.81	0.41612	10.64
	aspect + Z(year) + reef(year) + year	0.915	0.08	3610.81	0.41612	9.64

(continued)

Response	Model	Δ AICc	ω AICc	AICc	R^2	EDF
	roughness(year) + Z(year) + reef(year) + year	1.379	0.064	3611.28	0.42538	11.76
	roughness(year) + aspect + Z(year) + reef(year) + year	1.383	0.063	3611.28	0.42538	12.76
	roughness(year) + aspect(year) + Z(year) + reef(year) + year	1.386	0.063	3611.28	0.42539	13.76
Large Reef Fish Index*	**roughness(year) + aspect + Z(year) + reef(year) + year**	**0**	**0.377**	**5793.34**	**0.38378**	**10.79**
	roughness(year) + aspect(year) + Z(year) + reef(year) + year	1.086	0.219	5794.43	0.38166	11.63
	roughness + aspect + Z(year) + reef(year) + year	1.864	0.148	5795.20	0.37894	10.13
	roughness + aspect(year) + Z(year) + reef(year) + year	1.886	0.147	5795.23	0.37821	10.97
Reef Fish Thermal Index	**aspect(year) + Z(year) + reef(year) + year**	**0**	**0.148**	**852.86**	**0.25025**	**10.62**
	aspect + Z(year) + reef(year) + year	0.098	0.141	852.96	0.25028	9.68
	aspect + Z(year) + reef + year	1.859	0.059	854.72	0.24281	8.61
	aspect(year) + Z(year) + reef + year	1.978	0.055	854.84	0.24222	9.52

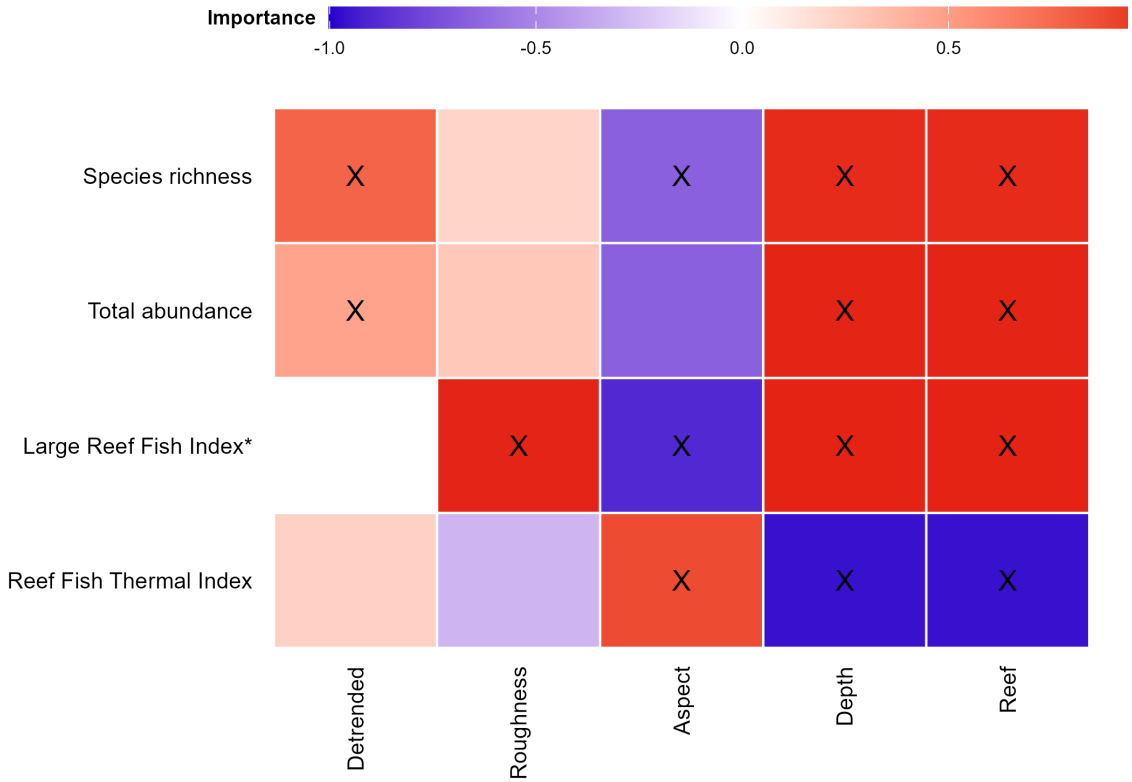


Figure D.4: Variable importance scores for predicting the probability of demersal fish metrics. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

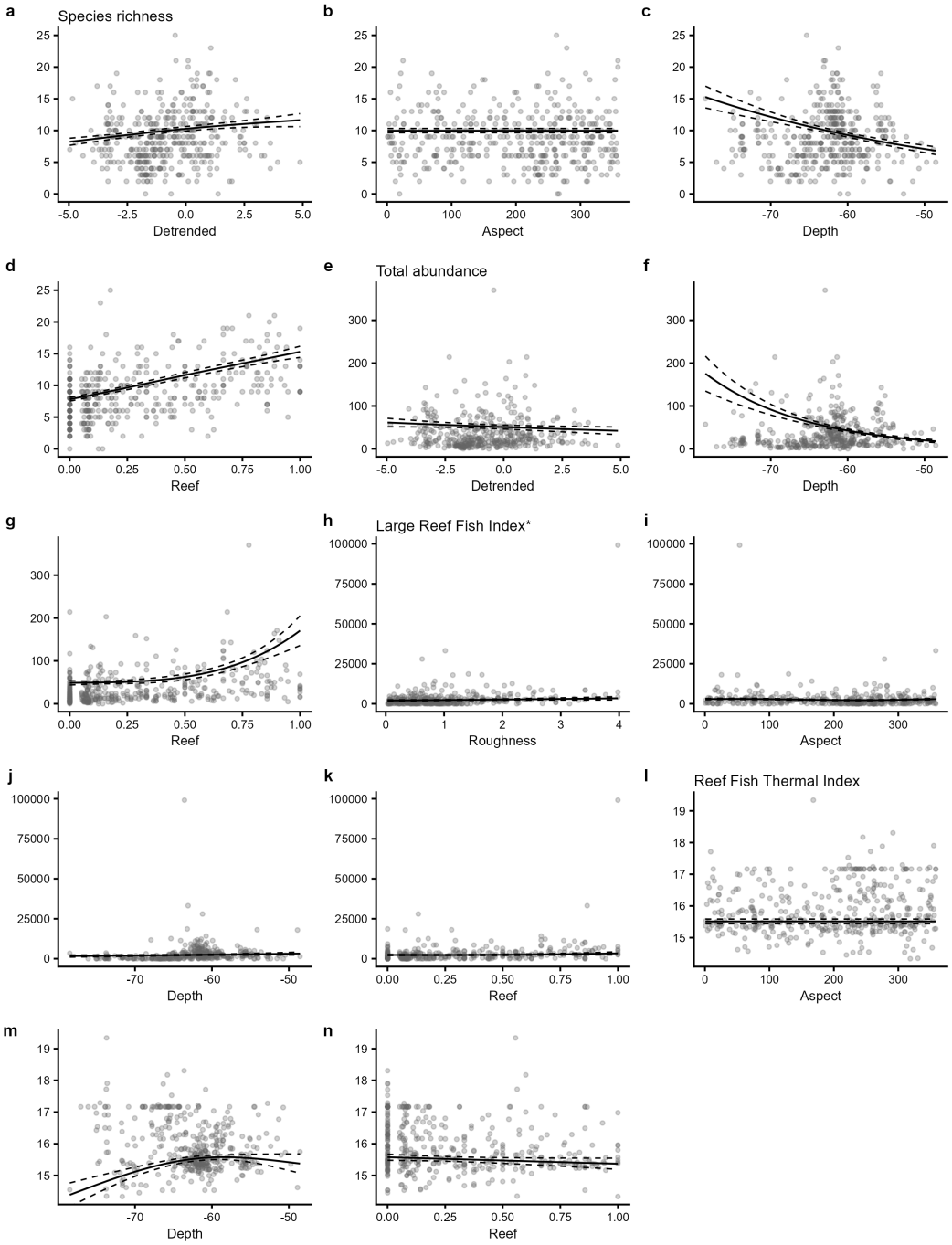


Figure D.5: Plots of the best model found to predict the probability of occurrence of key fish groups. Individual panes display the predicted relationships between model covariates and each demersal fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value, with year held at 2018 and zone status held at areas open to fishing. These terms are held only where model selection retained them: zone status was not retained for Species richness, Total abundance, Large Reef Fish Index and Reef Fish Thermal Index. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years and zones. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

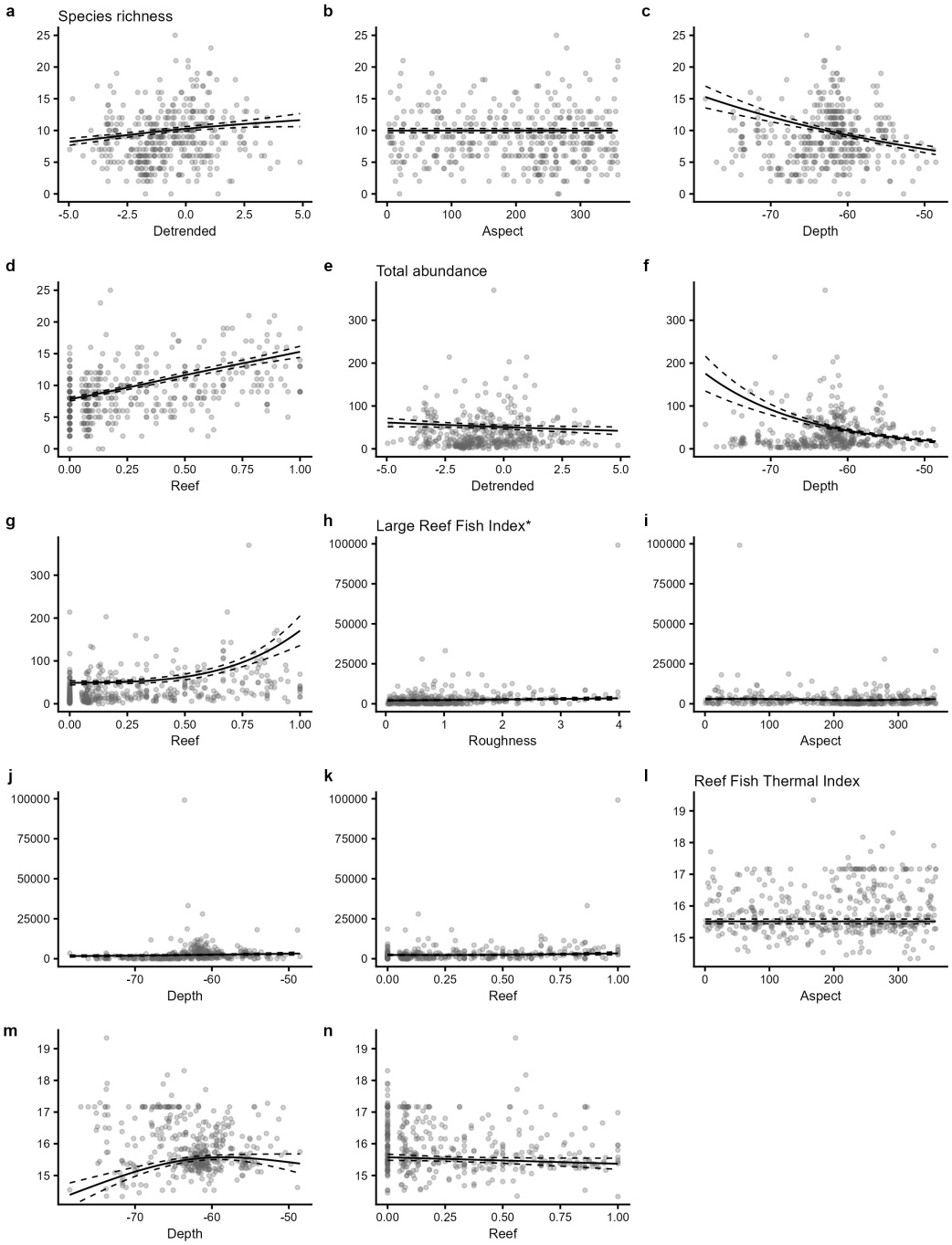


Figure D.6: Plots of the best model found to predict the probability of occurrence of key fish groups. Individual panes display the predicted relationships between model covariates and each demersal fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value, with year held at 2018 and zone status held at areas open to fishing. These terms are held only where model selection retained them: zone status was not retained for Species richness, Total abundance, Large Reef Fish Index and Reef Fish Thermal Index. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years and zones. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Significance of year and zone status

Table D.3: Significance of the year and zone status terms in each demersal fish GAMM. Estimate, standard error (SE), test statistic and p-value are taken from the parametric coefficient table of each model summary, one row per factor-level contrast (e.g. a survey year against the reference year, or No-Take against Fished).

Response	Term	Estimate	SE	Statistic	p-value
Species richness	Year: 2025 vs 2018	-0.157	0.036	-4.335	<0.001
Total abundance	Year: 2025 vs 2018	-0.890	0.074	-12.041	<0.001
Large Reef Fish Index*	Year: 2025 vs 2018	-0.293	0.122	-2.396	0.017
Reef Fish Thermal Index	Year: 2025 vs 2018	0.610	0.077	7.881	<0.001

Table D.4: Zone status test for each demersal fish metric, laid out by survey year. Zone status (No-Take vs Fished) is fitted as a single term across all years (no year by status interaction), so the status estimate, SE and p-value are the same in every row for a given metric; predicted values and their SE are shown per year, with other model covariates held at their mean.

Note
No fish models contained both year and status terms.